Locality of $\mathrm{Tr}(\phi^3)$ tree amplitudes from cyclic hidden zeros

1. Setup

Let $\{1, \ldots, j, \ldots, k, \ldots, n\}$ represent the n-ordered vertices of the n-gon. A *chord* is a variable X_{ij} with $1 \leq i < j \leq n$, $|j - i| \neq 1$ and $(j, i) \neq (1, n)$, thought of as the line between vertices i and j. The total chord count is $n(n-3)/2$. $X_{i,i+1}$ is not a variable, it is called a leg and is thought of as the side of the polygon connecting vertex i with $i + 1$

The most general rational ansatz of mass dimension $-2(n-3)$ with simple planar poles is

$$B_n \;=\; \sum_{|M|=n-3} \frac{a_M}{\prod_{X_c \in M} X_c},$$

where M ranges over multisets of $n-3$ chords. *Locality* is the statement that $a_M = 0$ whenever M contains two indices $(i, j), (r, s)$ with $1 \leq i < r < j < s \leq n$, i.e. their corresponding chords X_{ij} and X_{rs} intersect. If a set doesn't have intersecting chords, it is called a triangulation because the $n-3$ chords partition the n-gon into triangles. *Unitarity* is the statement that all triangulation coefficients are equal. Both should follow from the hidden zeros alone.

The hidden zero $\mathcal{Z}_{i,j}$. Given any short chord $X_{i,i+2}$we define the hidden zero surface $\mathcal{Z}_{ij}$ by:

$$X_{i+1,k} \;=\; X_{ik} - X_{i,i+2} \quad (k = 4, \ldots, n-1), \qquad X_{i+1,i+n-1} \;=\; -X_{i,i+2}. \tag{1}$$

The $n-3$ equations of (1) let us choose, for each $k \in \{4, \ldots, n-1\}$, either X_{ik} or $X_{i+1,k}$ to be the *free* variable; the other is then determined. Free variables often parametrize intersections and help force locality. The chord $X_{i+1,i+n-1}$ is always constrained, and it helps equate coefficients forcing unitarity. We call $X_{i,i+2}$ the *special* chord and $X_{i+1,i+n-1}$ the *bare* chord. Without loss of generality, by cyclic symmetry of hidden zeroes, we work in $\mathcal{Z}_{13}$, the linear subspace defined by:

$$X_{2k} \;=\; X_{1k} - X_{13} \quad (k = 4, \ldots, n-1), \qquad X_{2n} \;=\; -X_{13}. \tag{2}$$

By Rodina's lemma, if an amplitude vanishes on a hidden zero surface $\mathcal{Z}_{ij}$, then so does it's sub amplitudes. That is, if $B|_{\mathcal{Z}} = 0$ and $B \;=\; \sum_{n \geq 3} B_n$, then $B_n|_{\mathcal{Z}} = 0$, so we can work one n at a time.

The always-free chords. Chords with both endpoints in $\{3, 4, \ldots, n\}$ never appear in (2) at any n. Arranged by lower endpoint i and upper endpoint j:

	$j=5$	$j=6$	$j=7$	$\cdots$	$j=n$
$i=3$	X_{35}	X_{36}	X_{37}	$\cdots$	X_{3n}
$i=4$		X_{46}	X_{47}	$\cdots$	X_{4n}
$i=5$			X_{57}	$\cdots$	X_{5n}
$\vdots$				$\ddots$	$\vdots$
$i=n-2$					$X_{n-2,n}$

The triangular array has $\binom{n-3}{2}$ entries. Combined with one chord from each pair (X_{1k}, X_{2k}) for $k = 4, \ldots, n-1$ (our choice), they form the independent variables on $\mathcal{Z}_{1,3}$.

The choice function. For each $k \in \{4, \ldots, n-1\}$ pick $\sigma(k) \in \{1, 2\}$, declaring $X_{\sigma(k),k}$ to be the *constrained* variable in equation k of (2) and $X_{(3-\sigma(k)),k}$ the *free* one. Together with the always-free triangle, the set of free chords under σ is

$$\mathcal{F}_\sigma \;=\; \{X_{(3-\sigma(k)),k} : k = 4, \ldots, n-1\} \;\cup\; \{X_{ij} : 3 \le i < j \le n\}, \qquad |\mathcal{F}_\sigma| = \tfrac{(n-4)(n-1)}{2}.$$

The remaining $n-2$ chords — the special X_{13}, the bare X_{2n}, and the $n-4$ constrained companions $\{X_{\sigma(k),k}\}$ — are the constrained ones under σ.

2. The kill theorem

The setup of §1 fixes the zone $\mathcal{Z}_{1,3}$, the substitutions (2), the choice function $\sigma : \{4, \ldots, n-1\} \to \{1, 2\}$, and the free chord set

$$\mathcal{F}_\sigma \;=\; \{X_{(3-\sigma(k)),k} : k = 4, \ldots, n-1\} \;\cup\; \{X_{ij} : 3 \le i < j \le n\}.$$

Let Ω_σ denote the asymptotic limit

$$X_{13}, X_{2n} \to \infty \;\text{ and }\; X_{\sigma(k),k} \to \infty \;\text{ for each } k = 4, \ldots, n-1, \;\text{ all chords in } \mathcal{F}_\sigma \text{ held finite.}$$

Theorem 1 (Kill)**.** *Suppose $B_n\big|_{\mathcal{Z}_{1,3}} = 0$. Then for every multiset M of size $n-3$ with $M \subseteq \mathcal{F}_\sigma$, $a_M = 0$.*

Proof. The asymptotic limit Ω_σ sends constrained chords to infinity. A term $a_M / \prod_{c \in M} X_c$ survives this limit iff every chord in its denominator is a free chord under σ. Thus,

$$B_n\big|_{\Omega_\sigma} \;=\; \sum_{\substack{|M|=n-3 \\ M \subseteq \mathcal{F}_\sigma}} \frac{a_M}{\prod_{c \in M} X_c} \;=\; 0.$$

This identity is a Laurent polynomial in $|\mathcal{F}_\sigma|$ independent variables. All coefficients of a vanishing Laurent polynomial equal zero. Hence $a_M = 0$ for every $M \subseteq \mathcal{F}_\sigma$ with $|M| = n-3$. □

The kill set at $\mathcal{Z}_{1,3}$. A multiset M of size $n-3$ is killed at zone $\mathcal{Z}_{1,3}$ if there exists a choice function σ with $M \subseteq \mathcal{F}_\sigma$. Equivalently:

(K1) $X_{13} \notin M$ and $X_{2n} \notin M$;

(K2) for every $k \in \{4, \ldots, n-1\}$, M does not contain both X_{1k} and X_{2k}.

We say M is *killed* (without qualification) if it is killed at some zone — equivalently, if (K1)+(K2) hold for some cyclic rotation $\mathcal{Z}_{ij}$.

3. Triangulations are never killed

The kill theorem of §2 eliminates the coefficient a_M of every multiset M that fits inside some $\mathcal{F}_\sigma$ at some zone. We now show this never happens to a triangulation: for every triangulation T of the n-gon, there is no σ for which $T \subseteq \mathcal{F}_\sigma$. Equivalently, T violates (K1) or (K2) at every zone.

Theorem 2 (Triangulations escape)**.** *For every $n \ge 4$, every triangulation T of the n-gon, T is not killed at zone $\mathcal{Z}_{1,3}$.*

Proof. We show T violates (K1) or (K2) at $\mathcal{Z}_{1,3}$.

The leg X_{12} is a side of exactly one triangle of T (every polygon edge sits inside exactly one triangle in any triangulation). Let Δ be that triangle and let a be its third vertex, so $a \in \{3, 4, \ldots, n\}$. The two sides of Δ meeting at a are then $(1, a)$ and $(2, a)$, both of which are either legs or chords in T.

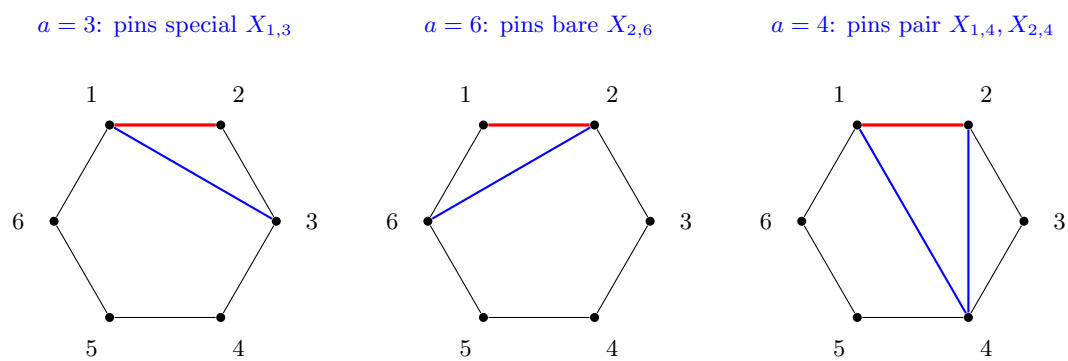
Case $a = 3$: $(1, a) = (1, 3)$ represent the special chord, so $X_{1,3} \in T$ and (K1) fails.

Case $a = n$: $(2, a) = (2, n)$ represent the bare chord, so $X_{2,n} \in T$ and (K1) fails.

Case $a \in \{4, \ldots, n-1\}$: both $(1, a)$ and $(2, a)$ are chords, neither is a leg. Both are in T, so the pair (X_{1a}, X_{2a}) is wholly in T, and (K2) fails at $k = a$.

These three cases exhaust the values of $a \in \{3, \ldots, n\}$, so T violates (K1) or (K2) at $\mathcal{Z}_{1,3}$. The substitutions (1) are cyclic shifts of (2), and the amplitude is cyclic-invariant, so the analysis at $\mathcal{Z}_{13}$ transfers to every zone $\mathcal{Z}_{r,r+2}$. □

Picture at $n = 6$. The proof is local to leg X_{12}. Whatever T looks like elsewhere, the unique triangle Δ of T containing $(1, 2)$ pins one of three things into T:



The red edge is the leg X_{12}; the blue chords are the sides of Δ that get pinned into T. In the leftmost figure $X_{1,3}$ (the special at $\mathcal{Z}_{1,3}$) is pinned; in the middle, $X_{2,6}$ (the bare); in the rightmost, both $X_{1,4}$ and $X_{2,4}$ (a constrained-companion pair). The vertex $a = 5$ would give the same conclusion as $a = 4$, with the pair $(X_{1,5}, X_{2,5})$ pinned.

4. Asymptotic completeness

The kill theorem of §2 eliminates every multiset that lies inside $\mathcal{F}_\sigma$ for some choice σ at some zone, and §3 shows no triangulation does. We now show that the kill mechanism kills *almost every* non-local coefficient as n grows.

Setting up the count. Let $N = n(n-3)/2$ be the number of chords. The total number of size-$(n-3)$ multisets of chords is

$$T(n) \;=\; \binom{N + n - 4}{n - 3}.$$

Let $S(n)$ be the number of multisets that survive at every zone — that is, multisets killed at no $\mathcal{Z}_{r,r+2}$. The non-triangulation survivors number $S(n) - C_{n-2}$, where $C_{n-2} = \frac{1}{n-1}\binom{2(n-2)}{n-2}$ is the Catalan count of triangulations.

Proposition 1 (Asymptotic completeness)**.**

$$\frac{S(n) - C_{n-2}}{T(n) - C_{n-2}} \longrightarrow 0 \qquad \textit{as } n \to \infty.$$

More precisely, $S(n)/T(n) \le 8/n + O(n^{-2})$.

The proof uses two elementary counting identities, then the union bound on the failure of (K1)+(K2) at a single zone.

Two counting identities. For a fixed chord X, the number of size-$(n-3)$ multisets containing X at multiplicity ≥ 1 is $\binom{N+n-5}{n-4}$, since one slot is fixed at X and the remaining $n-4$ slots range freely over all N chords. Therefore

$$\frac{\#\{M : X \in M\}}{T(n)} = \frac{\binom{N+n-5}{n-4}}{\binom{N+n-4}{n-3}} = \frac{n-3}{N+n-4}. \tag{3}$$

For two fixed distinct chords X, Y, the same argument with two slots fixed gives

$$\frac{\#\{M : X, Y \in M\}}{T(n)} = \frac{\binom{N+n-6}{n-5}}{\binom{N+n-4}{n-3}} = \frac{(n-3)(n-4)}{(N+n-4)(N+n-5)}. \tag{4}$$

These two identities use that the map $M \to M \setminus \{X\}$ is a bijection between multisets of size $n-3$ containing X and multisets of size $n-4$ over all N chords.

Proof of Proposition 1. Every survivor is not killed at zone 1, so

$$S(n) \le \big|\{M : M \text{ violates (K1) or (K2) at } \mathcal{Z}_{1,3}\}\big|.$$

A multiset M violates (K1) at $\mathcal{Z}_{1,3}$ iff $X_{1,3} \in M$ or $X_{2,n} \in M$. It violates (K2) iff for some $k \in \{4, \dots, n-1\}$, both $X_{1,k}$ and $X_{2,k}$ are in M. By the union bound,

$$\frac{S(n)}{T(n)} \le \frac{\#\{M : X_{1,3} \in M\}}{T(n)} + \frac{\#\{M : X_{2,n} \in M\}}{T(n)} + \sum_{k=4}^{n-1} \frac{\#\{M : X_{1,k}, X_{2,k} \in M\}}{T(n)}.$$

Apply (3) to the first two terms and (4) to each of the $n-4$ summands:

$$\frac{S(n)}{T(n)} \le \frac{2(n-3)}{N+n-4} + (n-4) \cdot \frac{(n-3)(n-4)}{(N+n-4)(N+n-5)}.$$

Using $N = n(n-3)/2$, the first term is $\frac{4(n-3)}{n^2-n-8} \sim 4/n$ and the second is $\frac{4(n-3)(n-4)^2}{(n^2-n-8)(n^2-n-10)} \sim 4/n$. Therefore

$$\frac{S(n)}{T(n)} \le \frac{8}{n} + O(n^{-2}) \longrightarrow 0.$$

Taking a moment to note that

$C_{n-2} \le 4^{n-2}$ and $T(n) = \binom{N+n-4}{n-3} \ge \left(\frac{N}{n-3}\right)^{n-3} = \left(\frac{n}{2}\right)^{n-3}$. Therefore $\frac{C_{n-2}}{T(n)} \le 4\left(\frac{8}{n}\right)^{n-3} \to 0$.

We conclude

$$\frac{S(n) - C_{n-2}}{T(n) - C_{n-2}} = \frac{S(n)/T(n) - C_{n-2}/T(n)}{1 - C_{n-2}/T(n)} \longrightarrow 0. \qquad \square$$

Numerical data. The asymptotic bound $S(n)/T(n) \le 8/n + O(n^{-2})$ is far from tight. The actual kill fractions among non-triangulation multisets, computed by exhaustive enumeration over all multisets and all zones, are:

n	5	6	7	8	9
$T(n)$	15	165	2,380	42,504	906,192
C_{n-2}	5	14	42	132	429
non-locals	10	151	2,338	42,372	905,763
survivors	0	0	7	100	1,011
kill %	100%	100%	99.70%	99.76%	99.89%

The empirical decay is much faster than $O(1/n)$. We do not prove a tighter bound here; whether $S(n)/T(n)$ decays exponentially in n is open.

5. The structure of Step-1 survivors and a Laurent kill

Step 1 leaves 7, 100, and 1,011 survivors at $n = 7, 8, 9$ respectively, all non-triangulations. We describe their structure and prove that the $n = 7$ survivors all vanish via a Laurent-expansion argument at one zone combined with Step 1 at a neighboring zone.

Two empirical regularities. Direct enumeration through $n = 9$ reveals that every Step-1 survivor satisfies:

(S1) *At least one short.* Every survivor multiset contains at least one short chord $X_{i,i+2}$.

(S2) *Every vertex is witnessed.* For every vertex $v \in \{1, \dots, n\}$, some chord of M either has v as an endpoint, or is the short chord $X_{v-1,v+1}$ enclosing v.

Proposition 2 (Vertex-witness theorem)**.** *If M is a Step-1 survivor of B_n, then every vertex of the n-gon is witnessed by M.*

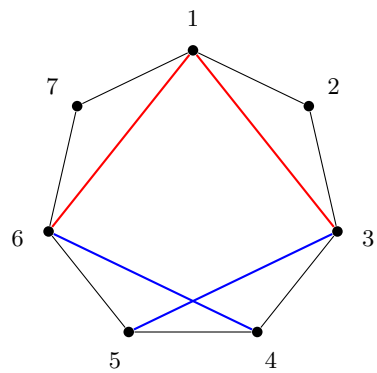
Proof. Suppose vertex v is not witnessed. Then no chord of M has v as an endpoint, and the short chord $X_{v-1,v+1}$ (which would witness v as an enclosed vertex) is not in M. We claim M is killed at zone $\mathcal{Z}_{v-1,v+1}$. The special at this zone is $X_{v-1,v+1} \notin M$ and the bare is incident to v hence $\notin M$, so (K1) holds. The constrained pairs at this zone are $(X_{v-1,k}, X_{v,k})$ for the appropriate range of k; each such pair has its second member incident to v, so it is not in M. Hence (K2) holds, and M is killed by Theorem 1 at zone $\mathcal{Z}_{v-1,v+1}$, contradicting that M survives. □

Property (S1) we leave as an empirical fact through $n = 9$. Heuristically, a short chord $X_{i,i+2}$ participates in two zones simultaneously — as the special at $\mathcal{Z}_{i,i+2}$ and as the bare at $\mathcal{Z}_{i+1,i+3}$ — so it offers an efficient way to violate (K1) at multiple zones. We do not have a proof that every survivor must contain one.

The fish at $n = 7$. The seven Step-1 survivors at $n = 7$ form a single cyclic orbit. The canonical representative is

$$M_\circ = \{X_{1,3},\, X_{1,6},\, X_{3,5},\, X_{4,6}\}.$$

The crossing $X_{3,5} \times X_{4,6}$ is the "tail" of the fish; the shorts $X_{1,3}$ and $X_{1,6}$ meeting at vertex 1 form the "head."



Theorem 3 (Fish vanishes)**.** *The fish coefficient* $a_{M_\circ} = 0$*, and by cyclic symmetry the coefficients of all seven* $n = 7$ *Step-1 survivors vanish.*